Growth and Decay in AI Memory

Growth and decay in artificial intelligence memory operate as coupled dynamical forces across three distinct substrates: parametric weights, latent activation states (such as key–value caches and recurrent hidden states), and token-level external stores. Across all three substrates, empirical research through late 2026 converges on a counterintuitive finding: unmanaged memory growth degrades model accuracy and system reliability long before storage limits are reached—manifesting as interference and address capture in weights, attention dilution in context windows, and memory bloat or embedding neighborhood shifts in external stores. Conversely, engineered decay mechanisms—including weight decay, retention gates, deferred cache eviction, and adaptive forgetting curves—often enable generalization, reasoning accuracy, and bounded latency, whereas naive time-based pruning or unguarded revocation can destroy critical constraints without ensuring true information removal.

Two evidentiary caveats frame current findings across these substrates. First, both parametric and external memory exhibit an access-versus-storage split: facts "forgotten" during sequential fine-tuning retain most of their underlying likelihood trace while losing retrieval access, and soft-revoked facts in production agent stores can leak back into prompts unless explicitly filtered at retrieval time. Second, quantitative evaluations of agentic memory growth and forgetting exhibit severe reproducibility gaps, with vendor-reported scores on LongMemEval and LoCoMo diverging sharply—by up to 45.4 percentage points in a third-party industry comparison (94.4% vendor-reported versus 49.0% in third-party/competitor testing)—due to proprietary platform optimizations, judge prompt differences, and model substitutions.

A Unified Framework Across Three Memory Substrates

Recent survey work, notably *Memory in the Age of AI Agents* (arXiv:2512.13564), replaces the traditional short-term versus long-term binary with a three-axis conceptual taxonomy spanning **form** (parametric, latent, and token-level), **function** (factual, experiential, and working), and **dynamics** (formation, evolution—consolidation, updating, and forgetting—and retrieval). Within this lifecycle, agent memory differs from static Retrieval-Augmented Generation (RAG) and raw context windows by persisting and continuously modifying state across interactions.

Substrate Form	Primary Storage Medium	Growth Mechanism	Empirical Scaling Ceiling or Threshold	Primary Decay Failure Mode	Engineered Decay or Regularization Mechanism
Parametric	Pretrained and fine-tuned weights	Gradient updates, SFT, RL, continual pretraining	~2.0–3.83 bits per parameter	Catastrophic forgetting (address capture) and plasticity loss	Weight decay, KL regularization, machine unlearning (GA, NPO, SimNPO, RMU)
Latent	Transformer KV caches, SSM	Autoregressive sequence	Effective length often 13–50%	Context rot, attention	KV eviction (DGE,

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	hidden states, test-time MLPs	expansion or online inner-loop updates	of nominal window on RULER (2024 models); sharp drop by 32K on NoLiMa	dilution, fixed-state copying limits	SnapKV), token merging, retention gates (G_t)
Token-Level	Discrete buffers, vector stores, temporal graphs, git-backed files	Turn extraction, ADD/UPDATE reconciliation, graph ingestion	Write-time compute/energy bottlenecks; vendor-reported BEAM drop from 1M to 10M tokens	Memory bloat, neighborhood shift, stale-fact retrieval leaks	Ebbinghaus, ACT-R, stretched exponential, and turn-count decay; temporal invalidation

Growth and decay behave inversely across these three substrates. Increasing parametric capacity raises the raw memorization ceiling linearly with parameter count, though excessive model scale under strong weight decay can trigger representation collapse in small-transformer grokking sweeps, and sustained sequential updates degrade network plasticity. Expanding transformer context windows raises maximum input limits to 400,000–1,000,000 tokens without preventing silent accuracy degradation inside the window. Expanding external token stores improves answer coverage while introducing candidate competition and embedding neighborhood shifts that erode retrieval precision.

Parametric Memory: Capacity Ceilings, Grokking, and Forgetting as Lost Access

Information-Theoretic Capacity and the Grokking Phase Transition

Parametric memory faces tight information-theoretic storage limits. Experiments by Morris et al. on GPT-2 architectures trained on uniform synthetic bitstrings measure maximum unintended memorization capacity at 3.51 bits per parameter in bfloat16 precision and 3.83 bits per parameter in fp32 precision—meaning that doubling numerical bit-width increases memorization density by roughly 9%. Earlier quantization-based estimates by Allen-Zhu and Li placed factual knowledge capacity near 2 bits per parameter, while a 2026 workshop study adapting Morris et al.'s random-label protocol reports a per-architecture capacity constant of approximately 2.16 bits per parameter in its central sweep, defining a threshold P_mem below which memorizing solutions cannot form. Crucially, Morris et al. observe that double descent begins precisely when dataset entropy exceeds total model capacity and unintended memorization ceases to lower training loss.

When overparameterized models learn algorithmic tasks, growth and decay interact across a staged phase transition known as **grokking**. Mechanistic interpretability work by Nanda et al. (2023) decomposes grokking into three sequential stages: early *memorization*, where training

loss collapses via high-norm lookup circuits; *circuit formation*, where structured generalizing circuits (such as Fourier multiplication representations) grow gradually alongside memorized circuits; and *cleanup*, where explicit weight decay prunes higher-norm memorizing weights, causing test loss to fall abruptly while Fourier sparsity (measured by the Gini coefficient) rises sharply. In this setting, parametric decay acts as the selection pressure that forces compact generalization.

Quantitative phase-diagram studies in 2026 clarify both the scaling laws and collapse boundaries of this decay pressure. Across 384 configurations of two-hidden-layer MLPs on modular arithmetic (arXiv:2609.10657v1), generalization onset time follows a power-law scaling relation:

$$T_{\text{grok}} \propto H^{-0.27}\, D^{-2.04}\, \eta^{-0.50}\, \lambda^{-0.64}$$

achieving $R^2 = 0.732$ (0.821 with interactions), where H is hidden width, D is data complexity/coverage, η is learning rate, and λ is weight decay. Data complexity/coverage ($D^{-2.04}$) outweighs hidden width ($H^{-0.27}$) by nearly an order of magnitude in exponent: under this fitted equation, doubling D reduces predicted T_{grok} by a factor of $2^{2.04} \approx 4.1\times$, compared with $2^{0.27} \approx 1.2\times$ for doubling H. Along the transition, weight norms compress to a median ratio $\|\theta(T_{grok})\| / \|\theta(T_{mem})\| = 0.42$. Critical weight-decay boundaries depend strongly on architecture and model size: while the two-layer MLP sweep identifies a lower threshold at $\lambda \geq 1.0$ below which generalization is suppressed, an independent researcher preprint evaluating 4-layer, 8-head transformers on mod-97 addition places the developmental threshold at $\lambda_c = 0.0158$ (95% CI [0.0109, 0.0200], N = 210), reporting 90–100% grokking over λ in [0.1, 2.0] (with onset falling from 1,090 to 83 epochs) alongside complete attention-head collapse at $\lambda = 10$ or at $\lambda = 1.0$ when scaling up to 85M parameters. Earlier effective theory by Liu et al. (2022) similarly linked grokking steps to the inverse of the slowest Hessian eigenvalue ($1 / (\lambda_3 \eta)$) above a critical training-data fraction ($r[span_93](start_span)[span_93](end_span)_c = 0.4$).

Continual Learning, Address Capture, and Plasticity Loss

Growing parametric memory sequentially after pretraining induces two competing pathologies: **catastrophic forgetting**, where new gradient updates erode previously acquired behavior, and **loss of plasticity**, established by Dohare et al. (*Nature*, 2024), where sustained sequential learning degrades a network's capacity to acquire new tasks.

Empirical research by O'Neill et al. (*Can a Language Model Learn Facts Continually in Its Weights?*, arXiv:2607.11020, revised October 2026) isolates the mechanics of continual factual failure across 4B and 8B models under supervised fine-tuning (SFT) and distillation. First, training data breadth governs factual usability rather than mere recitation: bare factual statements leave a 27.4-point accuracy gap between recitation and compositional use relative to in-context prompting, whereas diverse restatements ("study" training) shrink that gap to 5.4 points. Second, factual survival depends primarily on how *subsequent* updates are trained rather than how the initial fact was encoded. After 20 sequential writes, bare-statement facts retain 1% strict accuracy versus 46% for study facts (which plateau at 25–28% after 100 writes); across a full factorial sweep, switching later writes from repeated statements to varied restatements boosts earlier-fact retention by 37.6 percentage points (95% CI [28.9, 46.3]), whereas changing how the earlier fact itself was written yields a statistically negligible -2.4 points (95% CI [−9.2, 4.1]). Note, however, that community commentary on the study highlights an unisolated variable: repeated examples may induce larger net weight perturbations because update norms were not equalized across training regimes.

Most strikingly, O'Neill et al. provide evidence that behavioral forgetting destroys retrieval *access* rather than underlying *storage*. Facts failing every strict query after 20 writes still retain a median 69% (bare-statement) and 79% (study) of the log-probability lift created by their initial write—57% and 67% after drift correction—never reverting to the pre-write floor. In 68.8–70% of bare-statement failure cases, querying the *first* written entity causes the model to output content from the *twentieth* written fact (versus 1% under study training), a misrouting dynamic the authors term **address capture**. Consistent with address capture rather than trace erasure, retraining a forgotten study fact consumed the full step budget versus 8 steps for a never-written control fact, inverting classical relearning savings—though the authors caution this probe is suggestive rather than conclusive because only three study facts were forgotten in total. Collateral damage to general abilities tracks Kullback–Leibler (KL) divergence from the original model closely (correlation 0.83 across 12 conditions and 0.946[span_7](start_span)[span_7](end_span) in the factorial experiment).

Post-Training Paradigms and Structural Isolation

Optimization choice strongly mediates this stability–plasticity trade-off. Comparing post-training methods across arithmetic card games (GeneralPoints) and visual navigation (V-IRL) on Llama-3.2-Vision-11B, Chu et al. (ICML 2025) demonstrate that SFT memorizes training distributions and struggles out-of-distribution, whereas outcome-based reinforcement learning (RL) generalizes to unseen rules and visual environments—while still relying on SFT warmup to stabilize output formatting. Extending this insight to continual retention across Qwen 2.5 (3B, 7B, 14B) and robotics, Shenfeld et al. (*RL's Razor*, arXiv:2509.04259, 2025) find that on-policy RL incurs substantially less catastrophic forgetting than SFT at matched new-task accuracy because sampling from the model's own policy implicitly biases solutions toward minimal KL divergence from the base model. Plotting forgetting against new-task KL divergence aligns SFT and RL along a single predictive curve, and an artificially constructed KL-minimal "oracle SFT" matches RL's retention—indicating that KL proximity, rather than negative RL gradients alone, drives retention. OpenReview reviewers note that this KL relationship remains correlational without a neuron-level causal account and that observed LLM forgetting magnitudes were milder than in toy experiments.

Constraining parameter updates structurally provides a complementary buffer against decay. On Llama-2-7B, Biderman et al. (*LoRA Learns Less and Forgets Less*, TMLR 2024) show that full fine-tuning learns weight perturbations 10–100× higher in rank than standard Low-Rank Adaptation (LoRA). On code instruction fine-tuning (HumanEval pass@1), rank-16 and rank-64 LoRA trail full fine-tuning (0.358 and 0.417 after 4 epochs versus 0.497 after 8 epochs), though rank-256 closes the gap (0.498 at 4 epochs); on GSM8K continued pretraining, rank-256 LoRA reaches 0.187 at 8.6B tokens versus full fine-tuning's 0.230. In exchange, LoRA retains substantially more out-of-domain commonsense and science reasoning (average HellaSwag, ARC-Challenge, and WinoGrande score): at peak HumanEval in code continued pretraining, LoRA scores 0.64 versus 0.54 for full fine-tuning (and 0.617 versus 0.545 at 20B tokens for rank-256), and in code instruction tuning, LoRA retains 0.63 versus 0.45 for full fine-tuning while preserving higher output diversity. This advantage is domain-dependent: in math instruction tuning at epoch 4, LoRA retains 0.63 versus 0.57 for full fine-tuning, and high-rank LoRA narrows the retention advantage. To keep shared representations continuously trainable without cross-task collapse, the 2026 CRMA architecture (arXiv:2606.00382v1) pairs per-task LoRA adapters with a spectrally non-expansive ($\|g(x)\|_2 \leq \|x\|_2$) residual backbone adapter, eliminating the +42.96% ± 5.5 drift of naive shared LoRA on 5-domain Mistral-7B sequences and

outperforming frozen-substrate modular LoRA by 1.99% ± 0.54 in holdout loss across 3 seeds (an effect observed at 7B but absent at 1.1B, with LoRA-only versus LoRA+CRMA protocol parity left for follow-up).

Machine Unlearning and the Illusion of Parametric Erasure

Intentional parametric decay—**machine unlearning** for privacy, copyright, or safety—exhibits the inverse of continual learning's access failure: algorithms suppress surface generation without reliably erasing stored weights. Standard regularized formulations pair retain utility preservation with forget penalties such as Gradient Ascent (GA), Representation Misdirection for Unlearning (RMU, noted in prior comparisons as more stable than NPO), Negative Preference Optimization (NPO), or SimNPO (NeurIPS 2025, which eliminates NPO's reference-model bias via length-normalized gradients).
Yet evaluations on TOFU, MUSE, and WMDP expose pervasive fragility. In an ICLR 2025 study, low learning rates used to protect retain utility leave unlearned weights close enough to original values that post-hoc weight quantization snaps parameters back toward their pre-unlearning grid, catastrophically recovering "erased" knowledge. Similarly, an OpenReview benchmark study demonstrates that unlearned models—including parameter-efficient LoUK on Llama-3.2-1B—leak supposedly forgotten information once evaluated under higher-temperature probabilistic decoding rather than greedy decoding. Unlearned models also remain vulnerable to rapid relearning attacks, and a November 2026 evaluation of 11 unlearning methods across TOFU (Llama-3 1B/3B/8B) and MUSE (Llama-2-7B) finds that no tested algorithm simultaneously satisfies forget quality, retain utility, leakage robustness, and provider auditability without auxiliary output-filtering pipelines.

Latent and Working Memory: Context Rot, Eviction Timing, and Recurrent State

Attention Dilution and Context Rot

During Transformer inference, working memory grows linearly with token length. Although provider-advertised context windows reached 400,000 tokens (GPT-5) and 1,000,000 tokens (Claude Opus 5 and Sonnet 5) by 2026, empirical benchmarks demonstrate a sharp divergence between **context window overflow** (a hard truncation failure at the token limit) and **context rot** (silent accuracy degradation occurring well inside the advertised ceiling).

Study or Benchmark	Phenomenon Evaluated	Key Quantitative Finding	Effective Context Implication
Lost in the Middle (Stanford, 2023/2024)	Positional U-curve bias in multi-doc QA	Accuracy drops from ~75% (position 1 of 20) to ~55% (position 10)	Middle positions receive systematically less usable attention
RULER (NVIDIA, ACL Findings 2024, via secondary summaries)	Multi-hop tracing, aggregation, QA beyond single needles	85.6% threshold met at 13% of window for Gemini 1.5 Pro (128K/1M), 32% for Claude 3.5 Sonnet (64K/200K), 50% for	Advertised window size overstates reliable multi-step context, though 2025 models improved

Study or Benchmark	Phenomenon Evaluated	Key Quantitative Finding	Effective Context Implication
		GPT-4 Turbo (64K/128K), Llama 3.1 70B, and Mixtral 8x22B; 2025 Qwen3-32B and GLM4 hold past 128K	
NoLiMa (Adobe Research, ICML 2025, via secondary summaries)	Non-literal retrieval without lexical overlap cues	10–11 of 12–13 evaluated models drop >50% from <1K baseline by 32K tokens; GPT-4o falls from 99.3% to 69.7%	Absence of literal string cues collapses effective context length
Context Rot Study (Chroma, Jul 2025, via vendor/secondary summaries)	Constant-difficulty length and distractor degradation across 18 models	All 18 models (including GPT-4.1, Claude 4, Gemini 2.5, Qwen3) degrade non-uniformly; ~7.9% length-only floor before distractors; coherent haystacks score worse than shuffled ones	Increasing token count alone degrades accuracy regardless of task complexity
Attention-Mass Trial Study (arXiv:2611.04169, Nov 2026)	Causal isolation of sequence length vs. evidence attention mass (a*)	Across 126,473 trials (5 models, 8K–32K), log(1/a*) explains 35–58% of loss variance vs. 2–8% for log(length)	Models tolerate ~50% loss of evidence attention mass, then fail super-linearly

These datasets reconcile seemingly contradictory empirical claims. While single-needle literal recall can remain above 99% up to 10M tokens (as reported for Gemini 1.5), non-literal inference and distractor resistance break down far earlier. An industry analysis synthesizing RULER and Chroma data (llm-stats.com) suggests a practical working ceiling of roughly 25–30% of the nominal context window before length penalties and U-shaped positional troughs compound nonlinearly. Although Chroma's 2025 tests found reduced pure needle-position sensitivity in newer frontier models compared with 2023 architectures, the November 2026 large-trial decomposition confirms that **softmax attention dilution**—stretching finite probability mass across n^2 pairwise token interactions (10 billion pairs at 100K tokens) alongside attention sinks absorbing excess weight—is the dominant proximate cause of accuracy loss. Secondary summaries of Chroma's distractor tests also note asymmetric failure modes across model families, with Claude models skewing toward conservative abstention and GPT models showing higher hallucination rates. In multi-turn agents, context rot becomes self-reinforcing: degraded outputs prompt extra file reads and recovery retries, injecting stale tool traces that dilute attention further until the task fails.

KV-Cache Eviction and Timing-Driven Compression

Because uncompressed key–value (KV) caches consume GPU memory linearly with sequence length, serving frameworks rely on **KV-cache eviction** to actively decay transient latent state. Training-free heuristics—such as StreamingLLM (retaining initial attention sinks plus a recency window), H2O (cumulative attention accumulation), SnapKV (suffix observation-window scoring with max-pooling), and PyramidKV (layer-decaying budgets)—prune the cache immediately after prompt prefill. Token-merging methods, including EMS (Evict-then-Merge, adding 4.7% scalar-indicator overhead at head dimension 128), ChunkKV, and DynSplit-KV (Feb 2026, reporting a 2.2× speedup over FlashAttention and 2.6× peak memory reduction), attempt to retain semantic coverage rather than isolated tokens. However, simple post-RoPE key averaging loses roughly 30% of key norm to phase interference.

In 2026, **Draft-Guided Eviction (DGE)** (arXiv:2609.33334v1; OpenReview submission WGFgksL7DY) demonstrated that eviction *timing*, rather than token-scoring formulas, explains much of the quality collapse in prefill-time pruning. At a per-head budget B = 128, SnapKV misses an average 16.7% of actual decode attention mass (median 9.4%, 90th percentile 43%). Simply deferring eviction until after generating the first k = 2 answer tokens on the uncompressed cache—anchoring the decoding trajectory—lifts the LongBench average across six 3B–14B models from 35.3 (SnapKV) and 40.1 (LAQ) to 44.2, within 0.2 points of uncompressed FullKV, even when retaining SnapKV's unmodified window scorer (DGE-W). (At the specific B = 128 reproduction slice, baseline scores stand at SnapKV 42.2, PyramidKV 41.6, and StreamingLLM 36.5.) On Needle-in-a-Haystack (Qwen2.5-7B, B = 32), DGE (k = 2) scores 100.0 against FullKV's 98.9, SnapKV's 74.7, and H2O's 42.1, while cutting time-to-first-token to 0.93× SnapKV and amortizing one-off cache surgery overhead (1.45× at 32 tokens, versus 2.15× for H2O) to 1.02× by 512 output tokens. To eliminate explicit draft-generation latency altogether, Samsung Research's **LookaheadKV** (ICLR 2026) trains lightweight LoRA modules and learnable lookahead tokens to forecast future attention patterns directly during prefill.

Fixed-State Recurrence and Test-Time Neural Memory

State Space Models (SSMs) such as Mamba and Mamba-2 compress history into a fixed-size recurrent hidden state $h_t = A_t h_{t-1} + B_t x_t$ rather than an expanding KV cache. As established in *Repeat After Me* (Jelassi et al., ICML 2024) and formalized in 2026 regret analyses, this introduces a sharp architectural trade-off: Transformers maintain an exponential width advantage for exact copying and content-addressed lookup, whereas SSMs excel at short-range positional assembly and long-horizon belief maintenance via learned per-eigendirection exponential decay kernels. Interpretability probes on Mamba (EMNLP Findings 2026) also reveal a U-shaped recall curve: recent tokens reside in transient recurrent states (vulnerable to distractors), early items persist in a sparse subset of long-lived channels, and middle items fail once evicted from short memory without promotion.

To unite fixed-state efficiency with adaptive long-term recall, Behrouz et al. introduced **Titans** (NeurIPS 2025) alongside the unifying **MIRAS** (ICLR 2026) and **Nested Learning** frameworks. Titans equips sequence models with a deep MLP long-term memory module updated at test time via gradient descent on an associative key–value loss $\|M_{t-1}(k_t) - v_t\|_2^2$. Memory writes are modulated by a **surprise metric** combining momentary gradient magnitude with momentum, paired with an adaptive **weight-decay forgetting gate** $(1 - \alpha_t) M_{t-1}$. On S-NIAH-N at 16K tokens, Titans (Memory-as-Context) scores 97.4 versus 4.4 for Test-Time Training; in OpenReview rebuttal benchmarks on BABILong, Titans variants sustain 99 (2K), 94 (1M), and 69–79 (10M), broadly matching the Associative Recurrent Memory Transformer (98 at 2K, 93 at 1M, 77 at 10M). MIRAS generalizes these designs across memory architecture,

attentional bias, retention gate (G_t), and learning rule, explicitly reframing neural "forget gates" as **regularization penalties** that balance new adaptation against prior state retention (explored in Moneta, Yaad, and Memora via L_p/L_q objectives).
Independent scrutiny tempers these test-time memory results. Published Titans runs remain capped at 1.3B parameters without standardized FLOP or wallclock parity ablations against sparse-attention baselines. Moreover, an OpenReview audit of Titans-style test-time video memories emphasizes that minimizing forward-pass associative loss demonstrates online *optimization* on current inputs rather than guaranteed *retention* across subsequent interfering writes.

Token-Level Agentic Memory: Controlled Growth, Decay Curves, and Benchmark Realities

Growth Mechanics, Retrieval Crowding, and Lifecycle Cost

External token-level stores bypass fixed weight capacities and context limits, yet unconstrained appends produce "memory bloat" and **corpus distribution shift**. Practitioner production diagnostics (Towards AI) illustrate how adding 50,000 HR documents referencing employment "termination" alters embedding neighborhoods so that queries for legal "termination clauses" pull noisy HR matches even though zero legal embeddings changed. Vendor analyses attribute up to 73% of RAG failures to retrieval rather than generation (Atlan), citing *synthesis hallucinations* (where models invent causal links across separately retrieved facts) and *context poisoning* from conflicting document versions. Similarly, a vendor review (Onyx) contrasting ECIR 2026 and EMNLP 2025 retrieval scaling research emphasizes that larger corpora trade higher answer *coverage* against sharper candidate *competition*.

Framework / System	Storage & Growth Architecture	Update & Conflict Policy	Compute, Energy & Latency Profile	Default Decay & Revocation Behavior
Mem0 (OSS & Platform)	Vector facts (+ optional Mem0g graph in Pro/paper)	Paper/OSS: LLM ADD/UPDATE/DELETE/NOOP; Apr 2026 platform: append + read-time date ranking	<0.1 s read latency vs. ~38 s full-context (GPT-4.1-mini); 1,050 kJ construction/sample (4.02 h, 4,878 kJ suite total); $0.27 API / 132.02 MB RAM in cloud-edge eval	Decay opt-in (0.3×–1.5× rank bias); expiry-flag overrides can leak revoked facts at retrieval
Zep / Graphiti	Bi-temporal knowledge graph (valid_at/invalid_at, created_at/expired_at)	Closes invalid_at edge on contradiction while retaining episode history	Vendor reports 155–162 ms p95 read; $1.00 API / 876.25 MB peak cloud RAM in cloud-edge eval	Retains closed edges; default retrieval returns invalidated status without suppressing it
Letta (MemGPT)	Tiered blocks	Agent tool calls +	Highest MAB suite	Forgetting off by

Framework / System	Storage & Growth Architecture	Update & Conflict Policy	Compute, Energy & Latency Profile	Default Decay & Revocation Behavior
	(core, recall, archival) + git-tracked MemFS	background "sleep-time compute"	overhead: 14.36 h, 18,394 LLM calls, 15,429 kJ	default; compaction every 10 min
FadeMem (arXiv:2601.18642)	Dual-layer STM/LTM hierarchy with memory fusion	LLM-guided conflict resolution and consolidation	Reduces storage footprint by 45% (SRR = 0.45)	Stretched exponential (β = 1.2 STM, β = 0.8 LTM) + spacing reinforcement + pruning below ε_prune
BM25 / HippoRAG v2	Lexical index / OpenIE KG with Personalized PageRank	Static or batch index build without turn-by-turn LLM reconciliation	32 J (BM25) to 99 kJ (HippoRAG v2) construction/sample; 582 kJ (BM25) to 1,339[span_264](start_span)[span_264](end_span) kJ (HippoRAG v2) suite total	Stateless/static unless paired with external temporal guards

Two 2026 systems studies highlight underappreciated bottlenecks in token-level growth and revocation. First, a lifecycle cost audit on MemoryAgentBench (arXiv:2606.06448v2) reveals that write-time LLM extraction dominates total system energy: per-sample construction spans four orders of magnitude, from 32 J for BM25 and 99 kJ for HippoRAG v2 to 1,050 kJ for Mem0. Across the full evaluated workload, BM25 (47.0% accuracy, 16.3 min, 582 kJ) and HippoRAG v2 (44.3%, 44.2 min, 1,339 kJ) outperformed Mem0 (32.0%, 4.02 h, 4,878 kJ) and Letta (27.7%, 14.36 h, 15,429 kJ) on both accuracy and energy. Second, a September 2026 security evaluation (arXiv:2609.08258v1) demonstrates that **soft revocation fails at retrieval**: both Graphiti (by default) and Mem0 (under its expiry-flag override) expose validity metadata to the caller without filtering revoked records from search results, requiring an explicit application-layer stale-retrieval guard.

Forgetting Curves and Production Pruning Policies

Classical cache eviction policies are poorly suited to agent memory. As documented in technical implementation guides, fixed Time-to-Live (TTL) schedules and Least Recently Used (LRU) eviction treat rare but critical constraints—such as user allergies or compliance rules—as dead weight if unread for months, risking catastrophic policy violations. To distinguish transient noise from durable facts, research architectures formalize mathematical decay curves:

Decay Model / System	Mathematical Form	Core Temporal & Reinforcement Mechanics	Reported Quantitative Behavior
Exponential / MemoryBank (2023)	R(t) = exp(−t / S) (or 0.5^(t/h), where h = S	Initializes stability S = 1; on recall, resets t = 0	Modeled 7d / 14d / 30d retention: 0.819 →

Decay Model / System	Mathematical Form	Core Temporal & Reinforcement Mechanics	Reported Quantitative Behavior
	ln 2)	and increments S ← S + 1	0.549 → 0.247 (theagentfieldguide.com); 30d half-life equals S = 43.28 days
Generative Agents (2023)	recenc[span_286](start_span)[span_286](end_span)y = 0.995^(hours)	Min-max scaled and summed equally with LLM-scored importance and relevance	Triggers reflection when cumulative importance exceeds 150
MemoryOS (2025)	Heat = N_visit +[span_288](start_span)[span_288](end_span) L_interaction + exp(−Δt / μ)	Combines visit frequency, interaction length, and exponential recency (μ = 10^7 s)	Promotes segments from mid-term to long-term memory when Heat > τ (τ = 5)
ACT-R Power-Law Port	A = ln Σ t_j^(−0.5); P = 1 / [1 + exp((τ − A) / s)]	Sums power-law decays across every historical access age t_j, unifying recency and frequency	Modeled 7d / 14d / 30d recall (τ = −1, s = 0.4): 0.920 → 0.751 → 0.515, retaining heavy tails (theagentfieldguide.com)
FadeMem (arXiv:2601.18642, 2026)	v(t) = v_0 ex[span_297](start_span)[span_297](end_span)p[−λ (t − τ)^β]	Dual exponents: β = 1.2 (STM rapid fade), β = 0.8 (LTM slow fade); λ_base = 0.1 lowered by importance + spacing boost	29.43 multi-hop F1 on LoCoMo (vs. Mem0 28.37), 45% storage reduction, 85.9% factual consistency and 68.9% conflict-resolution accuracy on LTI-Bench
DMF (arXiv:2606.03463v1, Jun 2026)	t_1/2(Ω) = ln 2 / [λ (1 − η Ω)] over turn count Δn	Replaces wall-clock time with deterministic turn lag Δn = i_max − i and score inertia Ω (λ = 0.035, η = 0.5)	Computed from Eq. 19: half-life spans 19.8 turns (Ω = 0) to 39.6 turns (Ω = 1); keeps immutable raw text for query-time interpretation

In production practice, however, *The Agent Field Guide* reports that automated decay is disabled by default across major commercial memory platforms. Where enabled—such as Mem0 Platform's opt-in 0.3×–1.5× ranking bias across the last 20 accesses or Hindsight's 0.2-weight 365-day recency boost—forgetting functions exclusively as a soft retrieval-ranking penalty rather than destructive deletion. Raw records remain archived to support auditability and point-in-time ("as-of") queries, reserving hard deletion strictly for compliance and user erasure requests.

The Evaluation Reckoning: Vendor Disputes vs. Selective Forgetting

Failures

Quantitative evaluation of agentic growth and decay remains fragmented. Early conversational benchmarks—LoCoMo (~26K tokens per dialogue across ~600 turns, 1,986 QA pairs including 446 adversarial) and LongMemEval (500 questions across ~115K tokens in -S and ~1.5M tokens in -M)—primarily test static or single-update retrieval rather than multi-hop conflict supersession.

Benchmark & Source Context	System	Vendor-Reported / Self-Published Score	Alternative, Independent, or Stress-Test Score	Primary Driver of Gap or Failure
LongMemEval (vendor vs. third-party/competitor runs)	Mem0	93.4–94.4% (vendor managed platform, Apr 2026)	49.0% (competitor/third-party blog eval with GPT-4o); 66.88–68.44% (Zep eval)	Proprietary managed optimizations absent in OSS SDK; differing judge prompts
LongMemEval & LoCoMo (vendor vs. revised/competitor runs)	Zep / Graphiti	94.7% LoCoMo, 90.2% LongMemEval (May 2026, GPT-5.4)	75.14% ± 0.17 (revised after 9-pt calculation error); 71.2% LongMemEval (Mem0 harness); 63.8% LoCoMo (Dakera)	Scoring code revisions, upgraded reader/judge models (GPT-4o vs. GPT-5.4), token caps
BEAM (100K–10M tokens, vendor-published)	Mem0	64.1% at 1M tokens → 48.6% at 10M tokens	No Zep-published BEAM figures	Severe accuracy decay across 10× extreme scale growth
MemoryAgentBench — Selective Forgetting (ICLR 2026, 262K)	Mem0 / HippoRAG v2 / Long-context baselines	Mem0: 10.0% (18% SH); BM25: 25.5% (48% SH); HippoRAG v2: 29.5% (54% SH)	GPT-5-mini long-context baseline: 53.0% (78% SH, 28% MH); o4-mini drops 80% (6K) → 14% (32K) MH	External memory stores fail multi-hop counterfactual updates; reasoning models collapse with length
ForgetEval-Adv (arXiv:2606.15903, 385 adversarial cases)	MemPalace / LangMem / Lethe / Mem0	MemPalace: 40% (60/150) on recall-heavy Memora	MemPalace: 0.0%; LangMem: 62.9%; Lethe: 63.4%; Mem0: 68.3%; Lethe+LLM: 91.7%	Axis flip: recall-focused systems lack forgetting primitives; deterministic rules score 0% on compound facts

Newer 2026 benchmarks isolate selective forgetting and adversarial supersession directly. In *MemoryAgentBench* (ICLR 2026, arXiv:2507.05257; 2,071 questions across 103K–1.44M tokens), every tested external memory framework trailed a raw long-context GPT-5-mini

baseline overall (60.6%) and on 262K-token selective forgetting (53.0%, versus 29.5% for HippoRAG v2, 25.5% for BM25, and 10.0% for Mem0). Multi-hop reasoning over updated facts proved fragile even for reasoning models: o4-mini dropped from 80% accuracy at 6K tokens to 14% at 32K tokens, and GPT-4o fell from 28% to 10%. Similarly, *ForgetEval* (arXiv:2606.15903) documents a stark **axis flip**: MemPalace scores 40% on recall-heavy Memora yet fails 100% of adversarial forgetting cases (0/385). Even among systems with explicit deletion or update paths, Mem0 drops to 12/39 (31%) on prefix collisions (versus Lethe's 32/39 or 82%), Lethe fails 0/38 cross-lingual identifier cases (versus Mem0's 21/38 or 55%), and all deterministic non-LLM engines score 0/40 on compound-fact supersession—requiring an LLM mutation hook that lifts Lethe+LLM to 91.7% (353/385) at the cost of 2.3 s per mutation versus 74 ms deterministic.

Architectural Synthesis and Engineering Implications

Viewed across parametric, latent, and token-level research through late 2026, durable AI memory depends on assigning information to the substrate whose growth and decay physics match its lifecycle rather than treating memory as a monolithic store.

Because sequential parametric writes suffer address capture, 25–28% multi-write retention plateaus, and 27.4-point recitation-to-use gaps on bare facts, mutable world knowledge and compositional facts belong in token-level or structured graph context rather than model weights. Weight updates are better reserved for broad domain adaptation or behavioral alignment using KL-minimal on-policy RL, rank-calibrated adapters, or spectrally bounded residual backbones. Where privacy or compliance demands verifiable deletion, machine unlearning in weights cannot be treated as guaranteed erasure when weights remain exposed to quantization or stochastic sampling, making external storage isolation and access controls essential.

Within working context, prompt budgets must be engineered against effective attention rather than nominal window ceilings. Because super-linear accuracy loss emerges once evidence attention mass drops roughly in half, practical architectures—such as subagent isolation returning 50–200-token condensed summaries, deferred KV-cache eviction (DGE), boundary placement of instructions and provenance to avoid middle-trough decay, raw-record-grounded compaction rather than recursive summary-of-summary chains, and pinning immutable policy constraints outside the summarizer—preserve signal-to-noise ratio across long horizons.

Finally, external long-term stores should implement decay as reversible rank demotion and bi-temporal versioning rather than destructive TTL/LRU deletion or unguarded soft revocation. Combining explicit validity intervals (valid_at / invalid_at) or deterministic turn-count and power-law salience scores with an immutable raw archive, an enforced retrieval-time suppression guard for invalidated facts, and multi-axis benchmarking across write-time energy, 1M–10M scale curves, and multi-hop conflict supersession addresses the primary failure modes exposed across current AI memory systems.

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